

module-11-multiplication-near-base - Darshana (60-min workshop)

IKS Curriculum - primary band

Module 11 — Multiplication Near a Power of 10 (Darśana)

Sanskrit: *Nikhilam* · **Band:** Primary (3-5) · **Duration:** 60 min · **Path:** Darśana (Workshop)

Fast multiplication when numbers are near 10, 100, or 1000.

Darśana note. This workshop condenses Module 11's full 10-day arc into one hour. For the deeper version — daily lesson plans, quizzes, assessments, slide decks — switch to [Adhyayana](#) (the Full Course path).

Opening hook · 5 min

Write 9×8 on the board. “Quick answer? 72.” Now write 99×98 . “That’s harder. Today: a trick.”

Core idea · 15 min

For numbers JUST BELOW 100, use this trick:

99×98 : - 99 is $100 - 1$ (deficit 1) - 98 is $100 - 2$ (deficit 2) - Cross-subtract: $99 - 2 = 97$. (Same as $98 - 1 = 97$.) - Multiply deficits: $1 \times 2 = 2$. - Answer: 9702!

Chant: “Cross-subtract, multiply deficits!”

Demonstration · 10 min

Practice in chorus: 9×8 , 8×7 , 7×6 (single digit deficits from 10). Then 99×98 , 98×97 (deficits from 100).

Source moment · 5 min

Tell the story of an old number-trick from India that lets you multiply BIG numbers if they are close to 100.

Hands-on activity · 15 min

Worksheet: 8 problems of $NN \times NN$ where both numbers are 91-99. Pairs check.

Reflection + take-home · 10 min

Each child does one near-100 problem aloud. Cheer.

Go deeper into Adhyayana

The 10 days you're skipping if you only do this workshop:

- **Day 01** — Finger-trick prelude
- **Day 02** — 97×96 — two below base
- **Day 03** — 103×104 — two above base
- **Day 04** — Mixed cases
- **Day 05** — 3-digit near 1000
- **Day 06** — Algebra proof
- **Day 07** — When to use
- **Day 08** — Speed sprints
- **Day 09** — Mixed practice — combining $\times 11$ and Nikhilam
- **Day 10** — Final assessment

→ [Open the Adhyayana track for this module](#) to access all 10 days, the 4-audience PDFs, the slide deck, the quizzes, and the project rubric.